REPRESENTABILITY OF THE FUNCTOR OF POINTS WHERE A FINITELY PRESENTED MODULE IS FLAT IN DIMENSIONS AT LEAST D, GIVEN A COMPLETE DEVISSAGE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex), a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros; no macro was invented or redefined, and the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 05UF (sources/stacks-flat/pinned-original-source.tex line 7806).

Allowed standard prerequisites (editorial).

- The flatness-in-dimension-n situation, its equation, and the one-step and complete devissage definitions. Same chapter, flat.tex, Situation labelled situation-flat-dimension-n (source line 6311), Equation labelled equation-flat-dimension-n (source line 6325), Definition labelled definition-one-step-devissage (source line 780) and Definition labelled definition-complete-devissage (source line 1195), all invoked in the statement and at proof lines 7830-7836. Fix the objects, the functor F_d and the devissage data (Z_k, Y_k, i_k, pi_k, G_k, alpha_k) with the smoothness of Y_k over S and the closedness of i_k.
- The induction step making alpha_k universally injective after shrinking. Same chapter, Lemma labelled lemma-induction-step-fp (source line 3141), invoked at proof lines 7867-7869. Supplies that after replacing S by an affine neighbourhood of s the map alpha_1 is S-universally injective, which produces the displayed short exact sequence.
- The boundary case d_1 = d and the comparison of the two functors. Same chapter, Lemmas labelled lemma-compare-H-F (source line 7778) and lemma-free-at-generic-points-representable (source line 7631), invoked at proof lines 7888-7891. Supply the representability conclusion in the remaining boundary case; the induction and the functorial reductions are inside the counted span.

Source reference. Original definition 05MT: sources/stacks-flat/pinned-original-source.tex, source lines 6311–6311. Same-chapter pointer into the author environment situation at source line 6311; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original equation 05MU: sources/stacks-flat/pinned-original-source.tex, source lines 6325–6325. Same-chapter pointer into the author environment equation at source line 6325; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 05HH: sources/stacks-flat/pinned-original-source.tex, source lines 1195–1195. Same-chapter pointer into the author environment definition at source line 1195; invoked at proof line 7832 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 05H4: sources/stacks-flat/pinned-original-source.tex, source lines 780–780. Same-chapter pointer into the author environment definition at source line 780; invoked at proof line 7835 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 05IA: sources/stacks-flat/pinned-original-source.tex, source lines 3141–3141. Same-chapter pointer into the author environment lemma at source line 3141; invoked at proof line 7868 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 05UE: sources/stacks-flat/pinned-original-source.tex, source lines 7778–7778. Same-chapter pointer into the author environment lemma at source line 7778; invoked at proof line 7889 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 05UC: sources/stacks-flat/pinned-original-source.tex, source lines 7631–7631. Same-chapter pointer into the author environment lemma at source line 7631; invoked at proof line 7891 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 05MT. Let $s \in S$ let $d \geq 0$. Assume*

(1) *there exists a complete dévissage of $\mathcal{F}/X/S$ over some point $s \in S$,*
(2) *X is of finite presentation over S,*
(3) *$\mathcal{F}$ is an $\mathcal{O}_X$-module of finite presentation, and*
(4) *$\mathcal{F}$ is flat in dimensions $\geq d + 1$ over S.*

Then after possibly replacing S by an open neighbourhood of s the functor F_d (05MU) is representable by a monomorphism $Z_d \to S$ of finite presentation.

Proof. A preliminary remark is that X, S are affine schemes and that it suffices to prove F_d is representable by a monomorphism of finite presentation $Z_d \to S$ on the category of affine schemes over S. (Of course we do not require Z_d to be affine.) Hence throughout the proof of the lemma we work in the category of affine schemes over S.

Let $(Z_k, Y_k, i_k, \pi_k, \mathcal{G}_k, \alpha_k)_{k=1,\ldots,n}$ be a complete dévissage of $\mathcal{F}/X/S$ over s, see Definition 05HH. We will use induction on the length n of the dévissage. Recall that $Y_k \to S$ is smooth with geometrically irreducible fibres, see Definition 05H4. Let d_k be the relative dimension of Y_k over S. Recall that $i_{k,*}\mathcal{G}_k = \mathrm{Coker}(\alpha_k)$ and that i_k is a closed immersion. By the definitions referenced above we have $d_1 = \dim(\mathrm{Supp}(\mathcal{F}_s))$ and

$$d_k = \dim(\mathrm{Supp}(\mathrm{Coker}(\alpha_{k-1})_s)) = \dim(\mathrm{Supp}(\mathcal{G}_{k,s}))$$

for $k = 2, \ldots, n$. It follows that $d_1 > d_2 > \ldots > d_n \geq 0$ because α_k is an isomorphism in the generic point of $(Y_k)_s$.

Note that i_1 is a closed immersion and $\mathcal{F} = i_{1,*}\mathcal{G}_1$. Hence for any morphism of schemes $T \to S$ with T affine, we have $\mathcal{F}_T = i_{1,T,*}\mathcal{G}_{1,T}$ and $i_{1,T}$ is still a closed immersion of schemes over T. Thus $\mathcal{F}_T$ is flat in dimensions $\geq d$ over T if and only if $\mathcal{G}_{1,T}$ is flat in dimensions $\geq d$ over T. Because $\pi_1 : Z_1 \to Y_1$ is finite we see in the same manner that $\mathcal{G}_{1,T}$ is flat in dimensions $\geq d$ over T if and only if $\pi_{1,T,*}\mathcal{G}_{1,T}$ is flat in dimensions $\geq d$ over T. The same arguments work for "flat in dimensions $\geq d+1$" and we conclude in particular that $\pi_{1,*}\mathcal{G}_1$ is flat over S in dimensions $\geq d+1$ by our assumption on $\mathcal{F}$.

Suppose that $d_1 > d$. It follows from the discussion above that in particular $\pi_{1,*}\mathcal{G}_1$ is flat over S at the generic point of $(Y_1)_s$. By Lemma 05IA we may replace S by an affine neighbourhood of s and assume that α_1 is S-universally injective. Because α_1 is S-universally injective, for any morphism $T \to S$ with T affine, we have a short exact sequence

$$0 \to \mathcal{O}_{Y_{1,T}}^{\oplus r_1} \to \pi_{1,T,*}\mathcal{G}_{1,T} \to \mathrm{Coker}(\alpha_1)_T \to 0$$

and still the first arrow is T-universally injective. Hence the set of points of $(Y_1)_T$ where $\pi_{1,T,*}\mathcal{G}_{1,T}$ is flat over T is the same as the set of points of $(Y_1)_T$ where $\mathrm{Coker}(\alpha_1)_T$ is flat over S. In this way the question reduces to the sheaf $\mathrm{Coker}(\alpha_1)$ which has a complete dévissage of length $n-1$ and we win by induction.

If $d_1 < d$ then F_d is represented by S and we win.

The last case is the case $d_1 = d$. This case follows from a combination of Lemma 05UE and Lemma 05UC. □